

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

NET 316 - R

Networking ask-me-anything talk

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Agenda

1. Quick introduction
2. You decide the rest; start thinking about your networking questions now

Ask us anything (Networking related)

What AWS considers in the “networking” specialty:

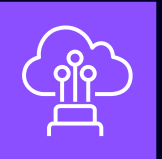


AWS Client VPN



AWS Site-to-Site VPN

Hybrid networking



AWS Direct Connect

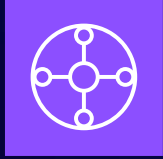


AWS Cloud WAN

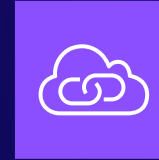


Amazon VPC

Core networking



AWS Transit Gateway



AWS PrivateLink

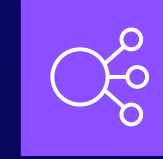


Amazon VPC Lattice

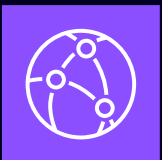
Application networking



Amazon Route 53



Elastic Load Balancing (ELB)



Amazon CloudFront

Edge



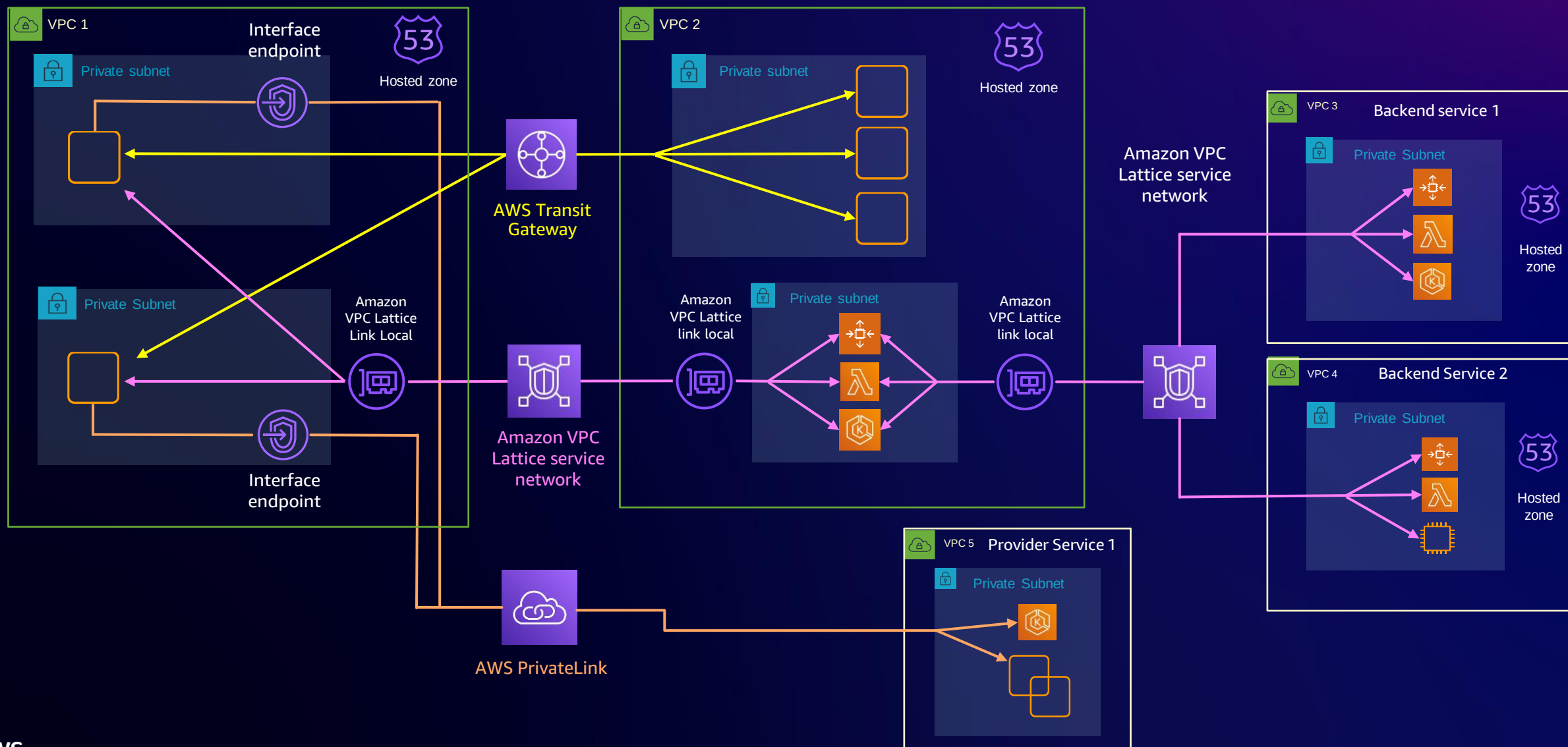
AWS Global Accelerator

Security



AWS Verified Access

Ask us about architectures



Doing this is easy

- 1 This is a question-and-answer session, with some drawing and hand-waving in between
- 2 If you can, start your question with what service or services you're going to ask about
- 3 If a line of questioning is too long, we will table it and meet after the session; we want everyone to have a chance to ask their question
 - We'll both be outside the room after this, and are at re:Invent all week – feel free to chat with us whenever!
 - There's also a networking specialty booth on the event floor – you can ask questions there as well



Question time



Thank you!



Please complete the session survey in the mobile app

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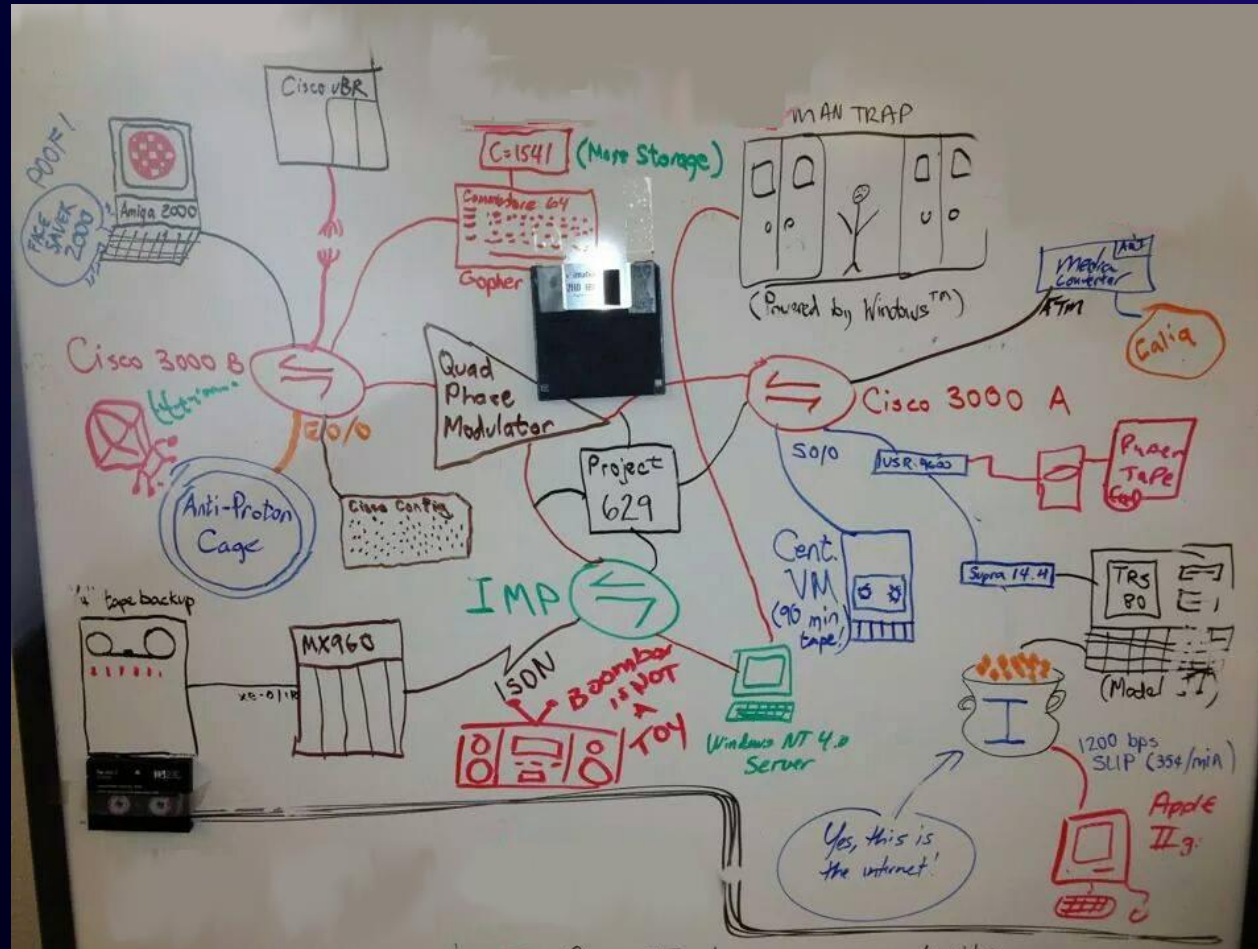
www.linkedin.com/in/james-m-wenzel



“I have two data centers – one on the US East Coast and one on the West. How should I set up Direct Connect for maximum resiliency and reliability?”

“My application developers are making all sorts of neat software, but the security group rules for allowing all the interconnections are getting difficult to maintain. Help!”

"I have this network diagram. What can I do to optimize?"

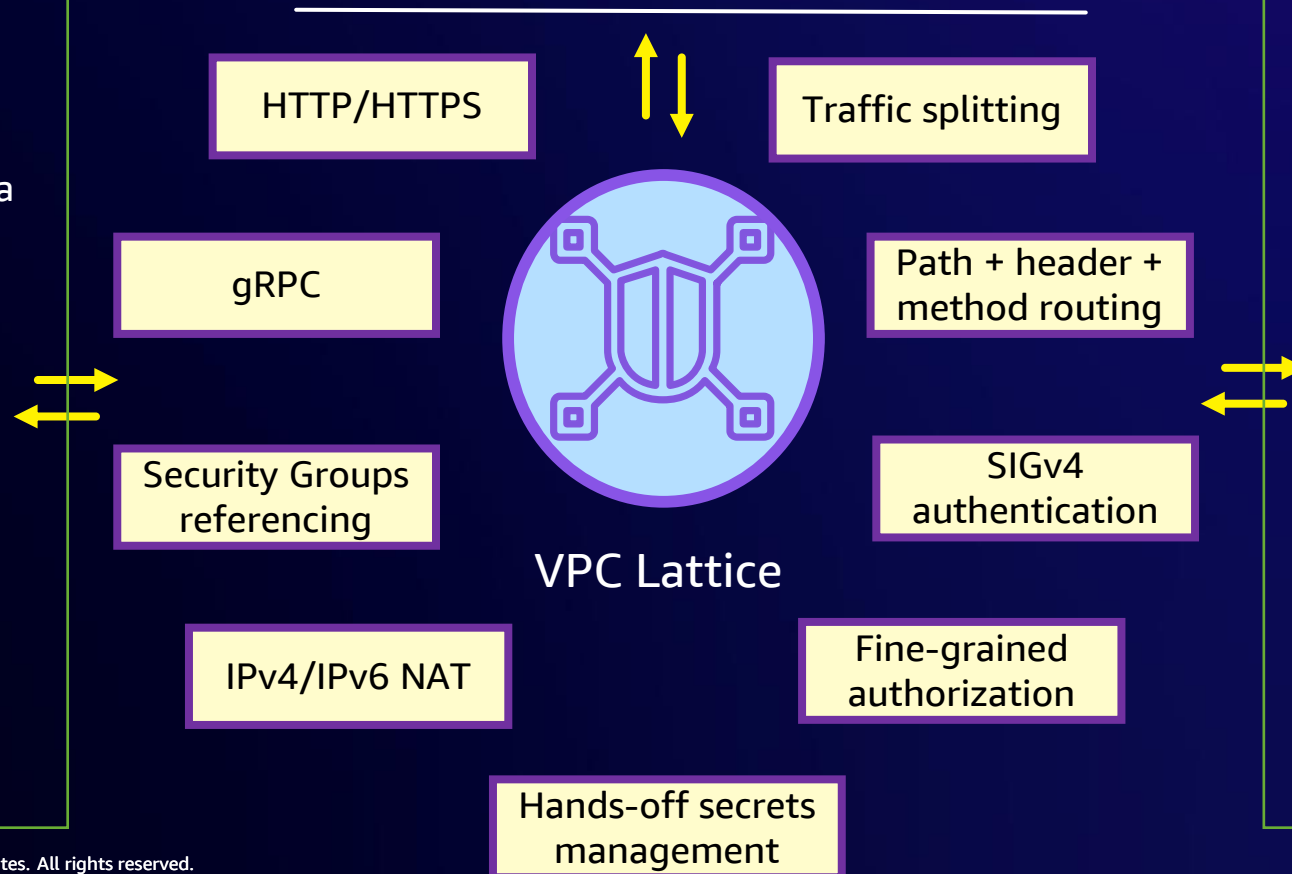
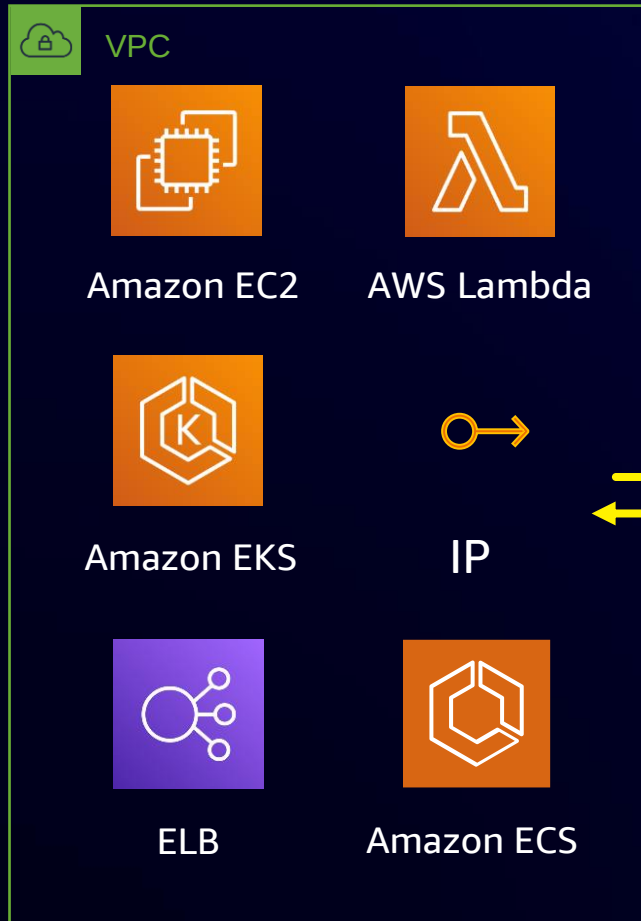


“I want to have my own firewall appliance inspect all traffic going between VPCs (‘east/west’) and going out (‘north/south’). What are some ideas on doing this?”

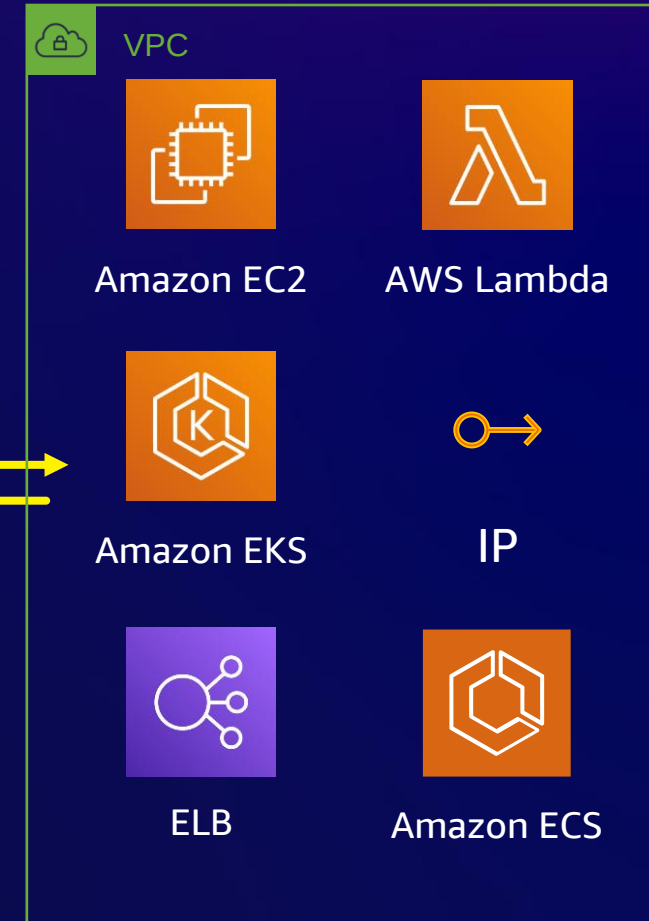
“I have an HTTPS application running inside AWS but need to scale it up. There’s Network Load Balancer, Application Load Balancer, and all sorts of TLS/SSL options. Help!”

Amazon VPC Lattice anatomy

Services

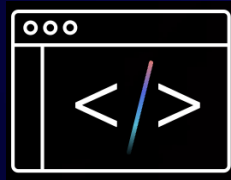
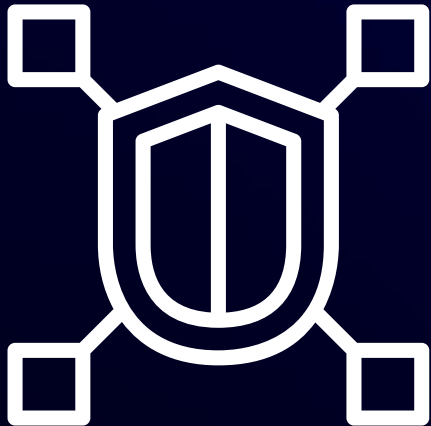


Services



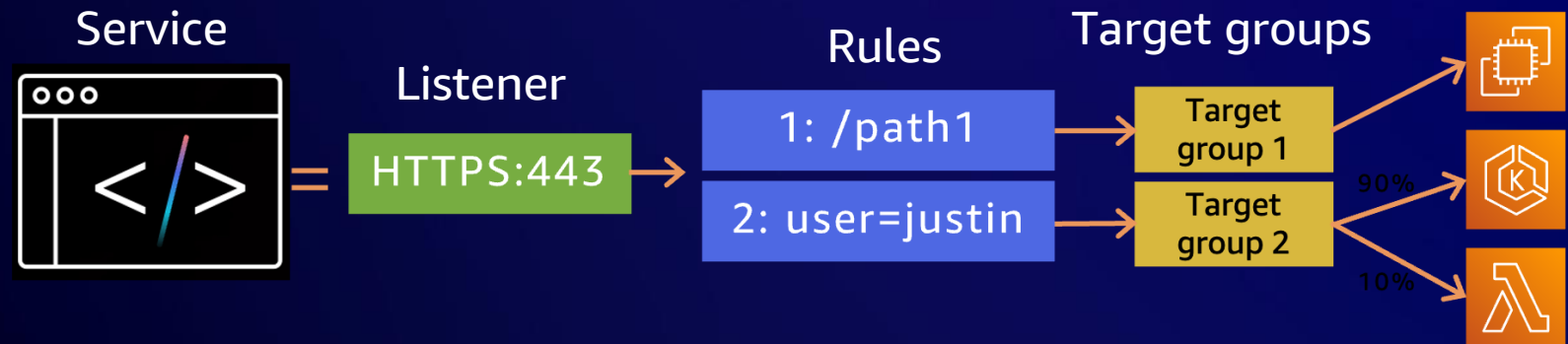
Anatomy of a Service

Amazon VPC Lattice

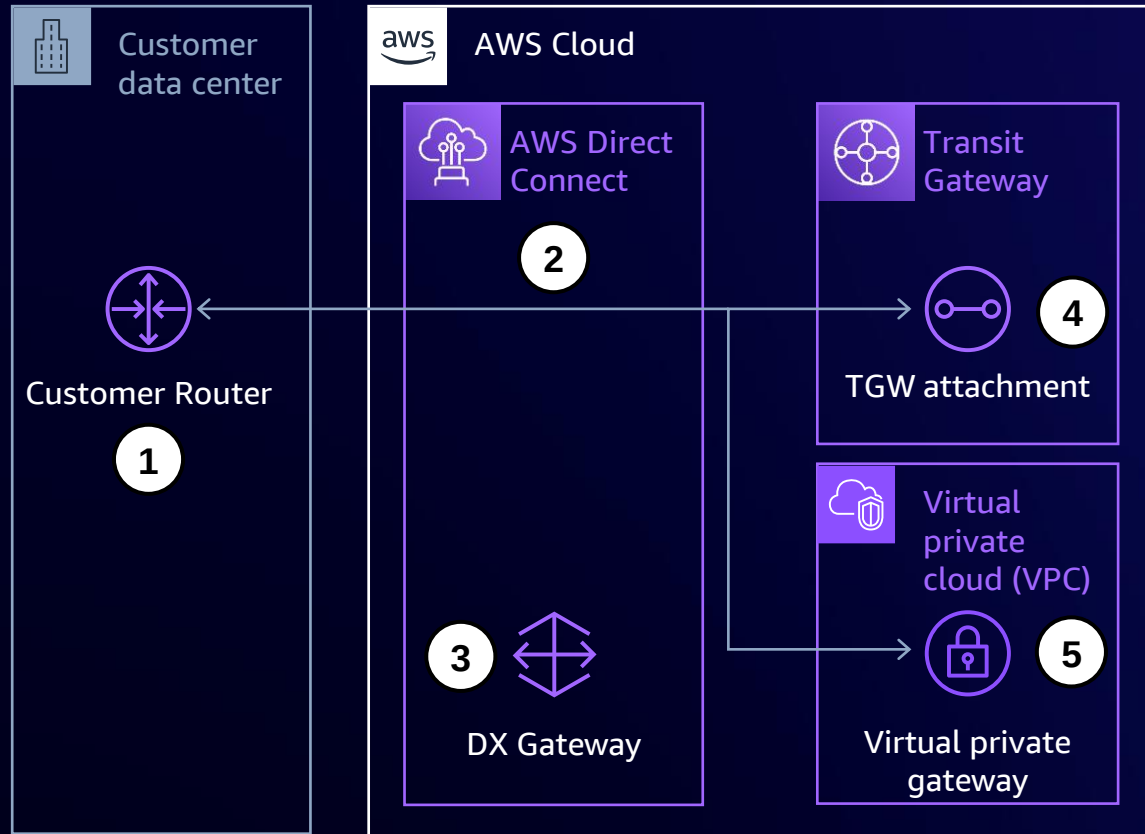


SERVICE

A unit of application running on instances, containers, and serverless and consisting of listeners, rules, and target groups



Monitoring AWS Direct Connect (DX)



Direct Connect key quotas:

Routes from customer to AWS, per BGP session, per AFI (private/transit): 100 ^(h)

Routes from customer to AWS, per BGP session, per AFI (public): 1000 ^(h)

Routes from Transit Gateway to customer, combined AFI: 200 ^(m)

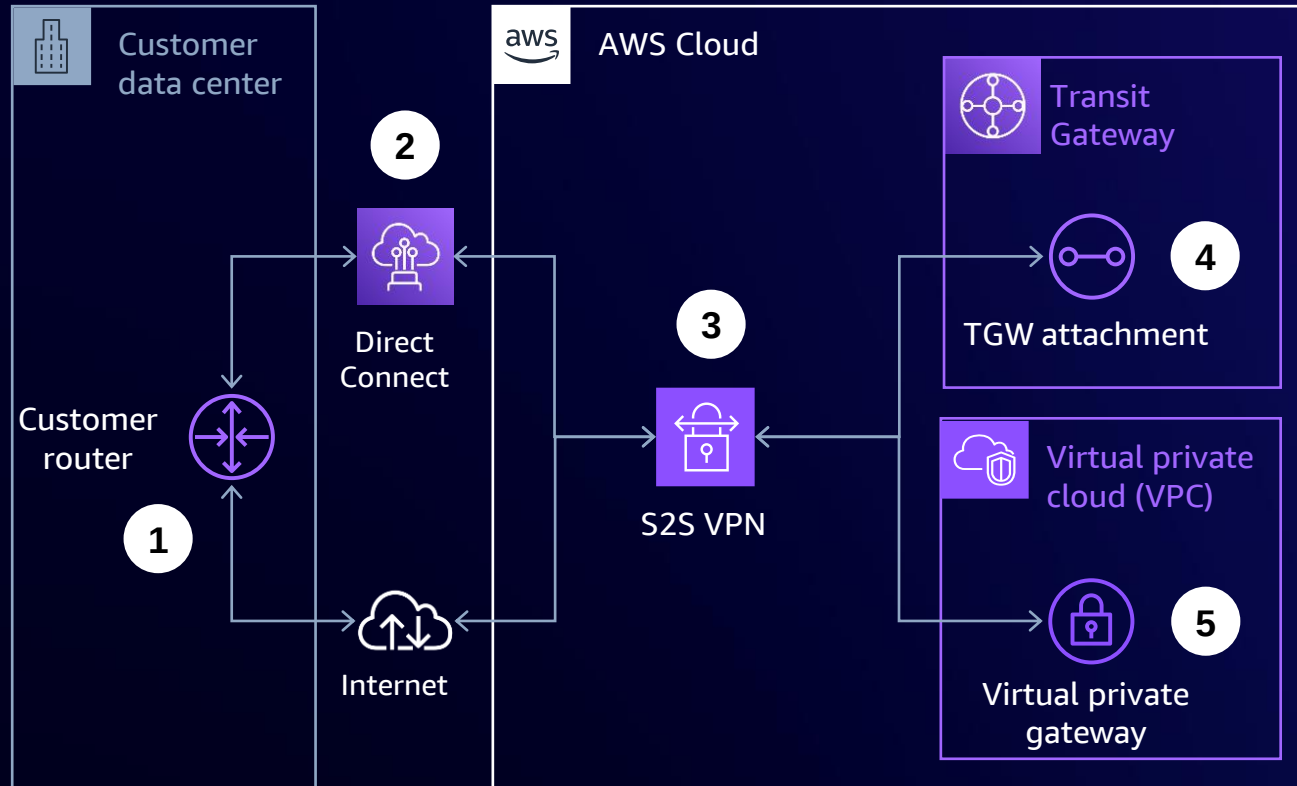
AWS Direct Connect metrics are in the AWS/DX CloudWatch namespace.

1. Monitor router incoming and outgoing BPS and PPS, along with number of routes advertised and received against quotas below
2. Alarm on `ConnectionState = 0` (link is down) and `ConnectionErrorCount` going up; monitor `Connection(Bps|Pps) (Ingress|Egress)` and alarm when at 80-90% of your connection's BPS; monitor `ConnectionLightLevelRx` and `ConnectionLightLevelTx` are staying between 0 and -10¹; you may wish to monitor the `VirtualInterface` version of these metrics per VIF to ensure one VIF isn't starving other VIFs
3. The DX Gateway does not sit in the traffic path and does not have separate monitoring
4. Monitor `(Packet|Bytes) DropCount (Blackhole|NoRoute)` if you have blackhole routes configured; monitor `(Bytes|Packets) (In|Out)` and compare against Customer Router side to verify counters on both sides are close
5. Virtual private gateway does not have metrics that can be monitored on itself – monitor the specific VIF metrics mentioned at point 2 instead

Documentation: <https://docs.aws.amazon.com/directconnect/latest/UserGuide/monitoring-cloudwatch.html>

¹ Some connections do not have these metrics available, and will show a flat 0.

Monitoring AWS Site-to-Site VPN (S2S VPN)



AWS Site-to-Site VPN metrics are in the `AWS/VPN` CloudWatch namespace.

1. Monitor router incoming and outgoing BPS and PPS, comparing against the numbers elsewhere; alarm on tunnel down or errors
2. See Direct Connect slide for DX monitoring details
3. Alarm on `TunnelState = 0` (down). Monitor/alarm on `TunnelDataIn + TunnelDataOut` against bandwidth quota
4. Monitor/Alarm TGW attachment
(Bytes|Packets) (In|Out) against S2S VPN quotas
5. Virtual Private Gateway does not have metrics that can be monitored on itself – monitor the S2S VPN instead

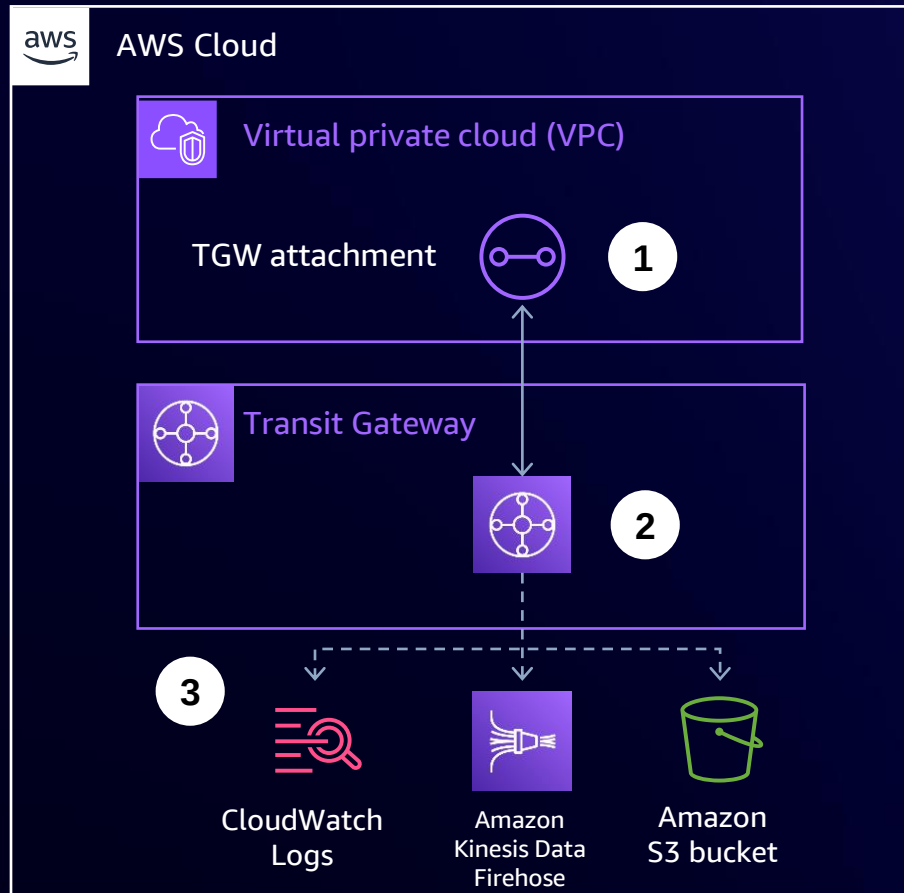
Site-to-Site VPN key quotas:

Bandwidth: 1.25 Gbps ^(h)

Packets per second: 140,000 ^(h)

Documentation: <https://docs.aws.amazon.com/vpn/latest/s2svpn/monitoring-cloudwatch-vpn.html>

Monitoring AWS Transit Gateway (TGW)



Transit Gateway metrics are in the `AWS/TransitGateway` CloudWatch namespace

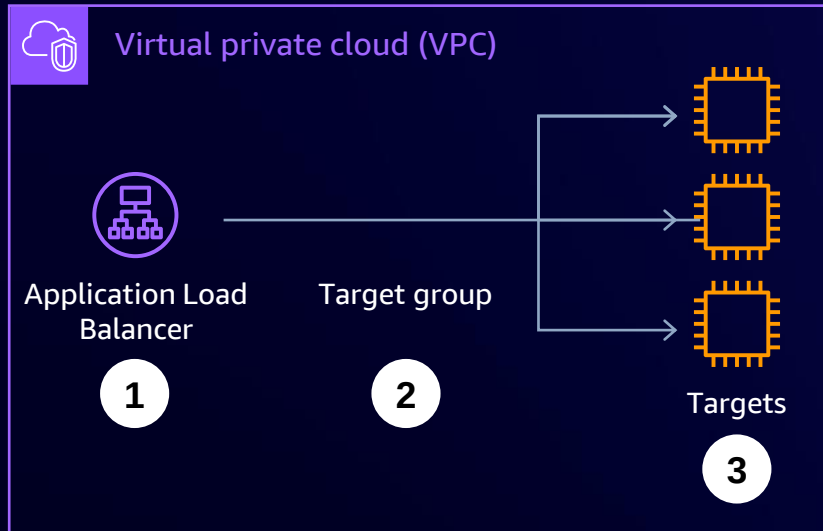
1. Monitor `(Packet|Bytes) DropCount (Blackhole|NoRoute)` if you have blackhole routes configured; monitor `(Bytes|Packets) (In|Out)` and compare against quotas below – alarm if approaching quotas
2. Monitor `(Packet|Bytes) DropCount (Blackhole|NoRoute)` if you have blackhole routes configured; monitor `(Bytes|Packets) (In|Out)` and compare against quotas below – alarm if approaching quotas
3. Consider enabling Transit Gateway Flow Logs
<https://docs.aws.amazon.com/vpc/latest/tgw/tgw-flow-logs.html>

Documentation: <https://docs.aws.amazon.com/vpc/latest/tgw/transit-gateway-cloudwatch-metrics.html>

Transit Gateway key quotas:

VPC, Direct Connect, and Peering Attachments, per AZ: 100G bps^(s), 7.5M pps^(s)

Monitoring Application Load Balancers (ALB)



Application Load Balancer metrics are in the `AWS/ApplicationELB` CloudWatch namespace.

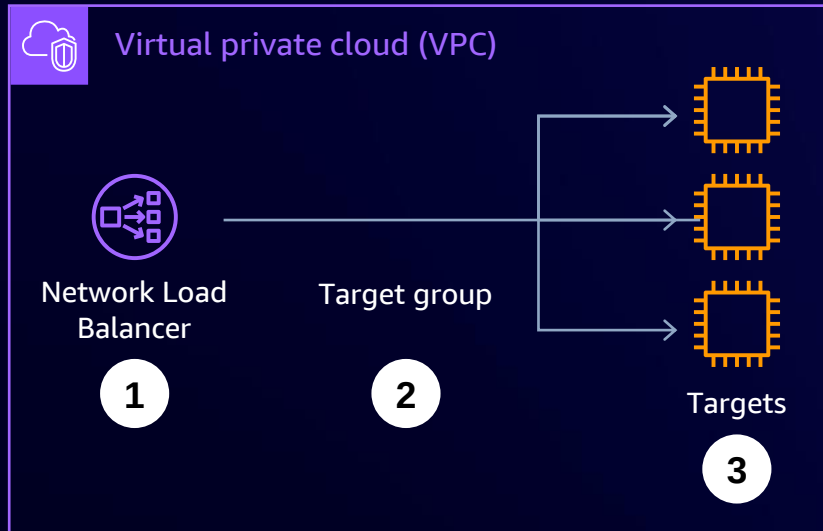
1. Monitor `ConsumedLCUs` as both a pricing metric and to watch for unexpected increases or drops; alarm on `RejectedConnectionCount` being non-zero – this indicates the ALB has reached its maximum number of connections. Monitor/alarm on `UnHealthyHostCount`, as appropriate; if using user authentication, monitor/alarm on `ELBAuthError` and for `ELBAuthLatency` going unusually high; monitor `ActiveConnectionCount`, `NewConnectionCount`, `ProcessedBytes`, and `ProcessedPackets` against planning guidelines
2. Monitor/alarm `UnhealthyRoutingRequestCount`, `UnhealthyStateDNS`, and `UnhealthyStateRouting`
3. Perform appropriate per-target monitoring; for instances, see instances page later in this deck

Documentation: <https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-cloudwatch-metrics.html>

ALB capacity planning guidelines:

The number of AZs in an ALB do not impact planning in terms of capacity

Monitoring Network Load Balancers (NLB)



Network Load Balancer metrics are in the `AWS/NetworkELB` CloudWatch namespace

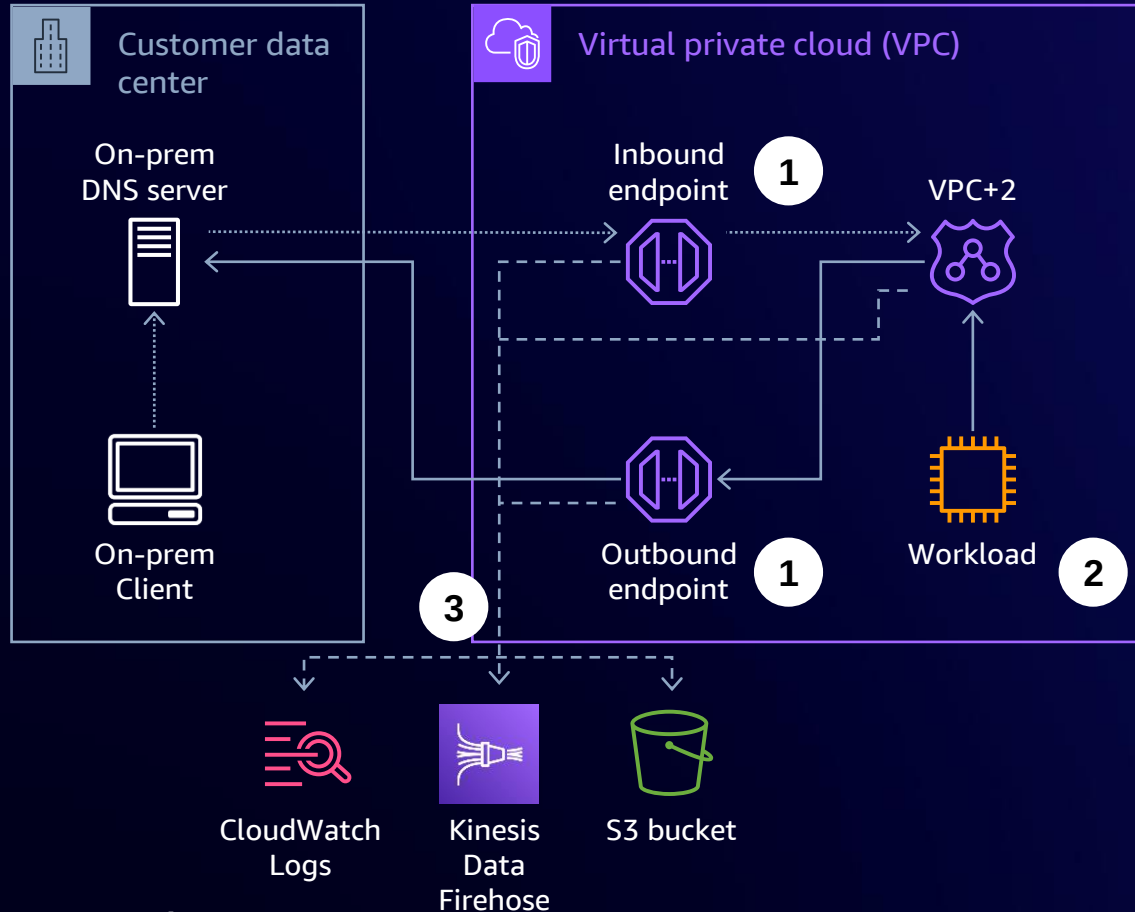
1. Monitor `ConsumedLCUs` as both a pricing metric and to watch for unexpected increases or drops; alarm on `PortAllocationErrorCount` being non-zero – this indicates the NLB has reached its maximum number of connections to a target; monitor/alarm on `UnHealthyHostCount` and `UnhealthyRoutingFlowCount`; monitor/alarm on `TCP_(Client|ELB|Target)_Reset_Count` if appropriate; monitor `ActiveFlowCount`, `NewFlowCount`, `ProcessedBytes` and `ProcessedPackets` against planning guidelines
2. Monitor/alarm `UnhealthyRoutingRequestCount`, `UnhealthyStateDNS`, and `UnhealthyStateRouting`
3. Perform appropriate per-target monitoring; for instances, see instances page later in this deck

Documentation: <https://docs.aws.amazon.com/elasticloadbalancing/latest/network/load-balancer-cloudwatch-metrics.html>

NLB capacity planning guidelines:

Unlike ALB, NLB capacities are per AZ, so you can scale by adding more AZs or NLBs

Monitoring Route 53 and Route 53 Resolver



Route53 key quotas:

Packets per second from an instance to link-local/VPC+2, per interface: 1,024 pps ^(h)

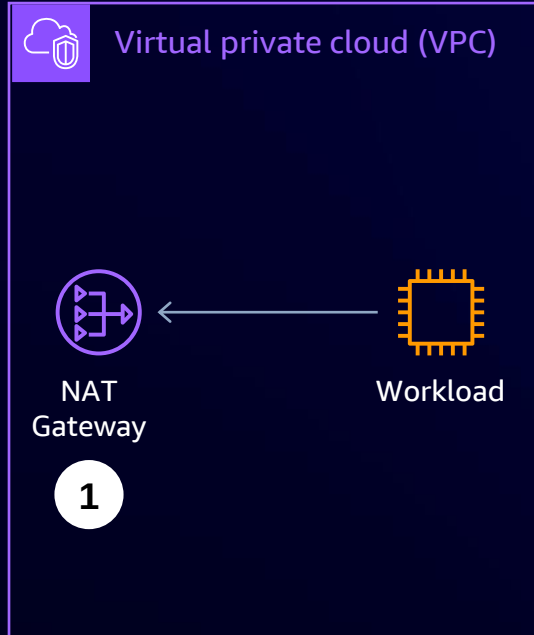
Queries per second per IP address in an endpoint: 10,000 qps ^(h)

Route 53 Resolver metrics are in the `AWS/Route53Resolver` CloudWatch namespace

1. Monitor `EndpointUnHealthyENICount` for alerts that resolver capacity is temporarily reduced; monitor/alarm `InboundQueryVolume` and `OutboundQueryAggregateVolume` per endpoint and per IP address against quotas; endpoints will be adding conntrack related metrics soon
2. Monitor instances' `linklocal_allowance_exceeded` network metric for traffic exceeding the 1,024 PPS per network interface limit
3. Consider enabling query logging (<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/resolver-query-logs.html>)

Documentation: <https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/monitoring-resolver-with-cloudwatch.html>

Monitoring NAT Gateway



NAT Gateway metrics are in the `AWS/NATGateway` CloudWatch namespace

1. Alarm on `PacketsDropCount` or `ErrorPortAllocation` errors rising quickly (more than a few per second); monitor that `BytesInFromSource+BytesInFromDestination` and `PacketsInFromSource+PacketsInFromDestination` remain within quotas listed below

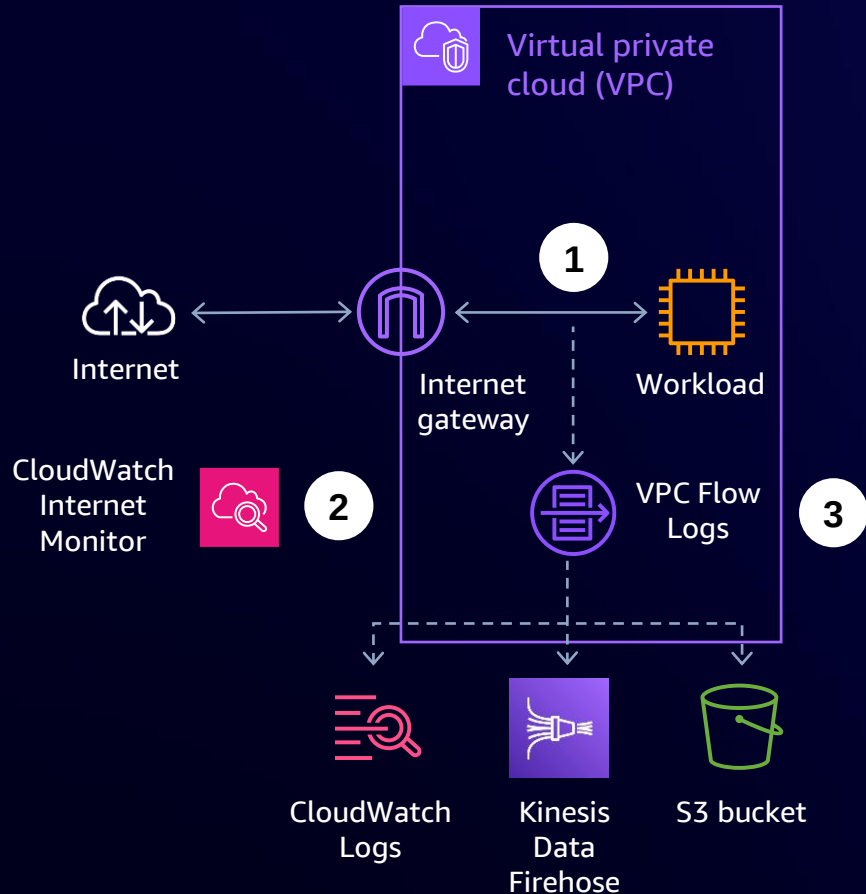
NAT Gateway key quotas:

Bits per second: 5 Gbps cold, able to scale up to 100 Gbps ^(h)

Simultaneous connections from a source IP to each unique destination: 55,000 ^(h) – can be mitigated by adding additional public IPs to the NAT gateway

Number of public IPs per NAT gateway: 2 ^(s, 8)

Monitoring internet gateway

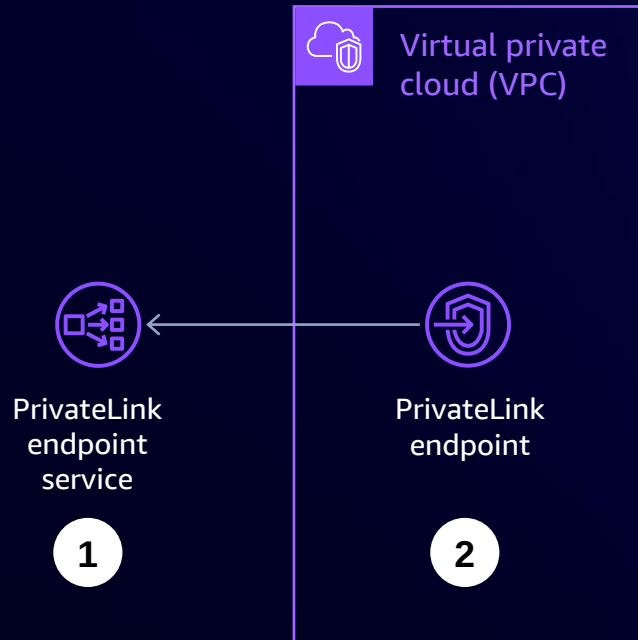


1. Monitor instances (see instances page) for network traffic limits exceeded
2. Consider enabling CloudWatch Internet Monitor:
<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/CloudWatch-InternetMonitor.html>
3. Consider enabling VPC Flow Logs:
<https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs.html>

Internet gateway key quotas:

There are no limits on internet gateway itself, however, individual instances are limited to either 5 Gbps ^(h), or for current generation instances with more than 32 vCPUs, 50% of their network bandwidth ^(h)

Monitoring AWS PrivateLink



1. Monitor endpoint services' (namespace `AWS/PrivateLinkServices`)
`ActiveConnections`, `BytesProcessed` and `NewConnections` against quotas below; alarm on `RstPacketsSent` increasing quickly; possibly monitor `EndpointsCount` metric for new endpoints connecting to a service
2. Monitor endpoints' (namespace `AWS/PrivateLinkEndpoints`)
`ActiveConnections`, `BytesProcessed` and `NewConnections` against quotas below; alarm on `PacketsDropped` and `RstPacketsReceived` increasing quickly; note that service interface endpoints are implemented using PrivateLink, and can be monitored the same way

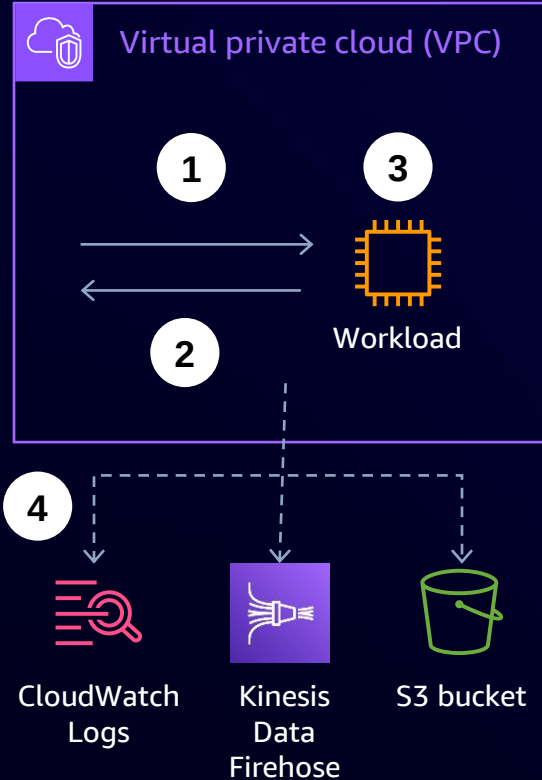
Documentation: <https://docs.aws.amazon.com/vpc/latest/privatelink/privatelink-cloudwatch-metrics.html>

PrivateLink key quotas:

Bits per second: 10 Gbps cold, able to scale up to 100 Gbps per AZ ^(h)

MTU: 8500 bytes ^(h)

Instances



It is strongly recommended instances install the Amazon CloudWatch agent, and export detailed networking statistics for the below numbers; that configuration defines the CloudWatch namespace; ensure the ENA driver on instances is active and up-to-date

1. For inbound network traffic, alarm on the `bw_in_allowance_exceeded` from `ethtool -S <interface>`, or interface RX dropped count from `ifconfig` rising quickly (more than a few per second)
2. For outbound network traffic, alarm on the `bw_out_allowance_exceeded` or `queue_<X>_tx_queue_stop` numbers from `ethtool -S <interface>`, or the dropped numbers from `tc -s qdisc show <interface>` rising quickly (more than a few per second)
3. In general, alarm on `pps_allowance_exceeded`, `conntrack_allowance_exceeded`, or `linklocal_allowance_exceeded`¹ increasing quickly; monitor the `conntrack_allowance_available` number from `ethtool -S <interface>` approaching zero, along with instance CPU and RAM usage percentages
4. Consider enabling VPC Flow Logs (<https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs.html>)

Documentation: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/viewing_metrics_with_cloudwatch.html
<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/monitoring-network-performance-ena.html>

Instances key quotas:

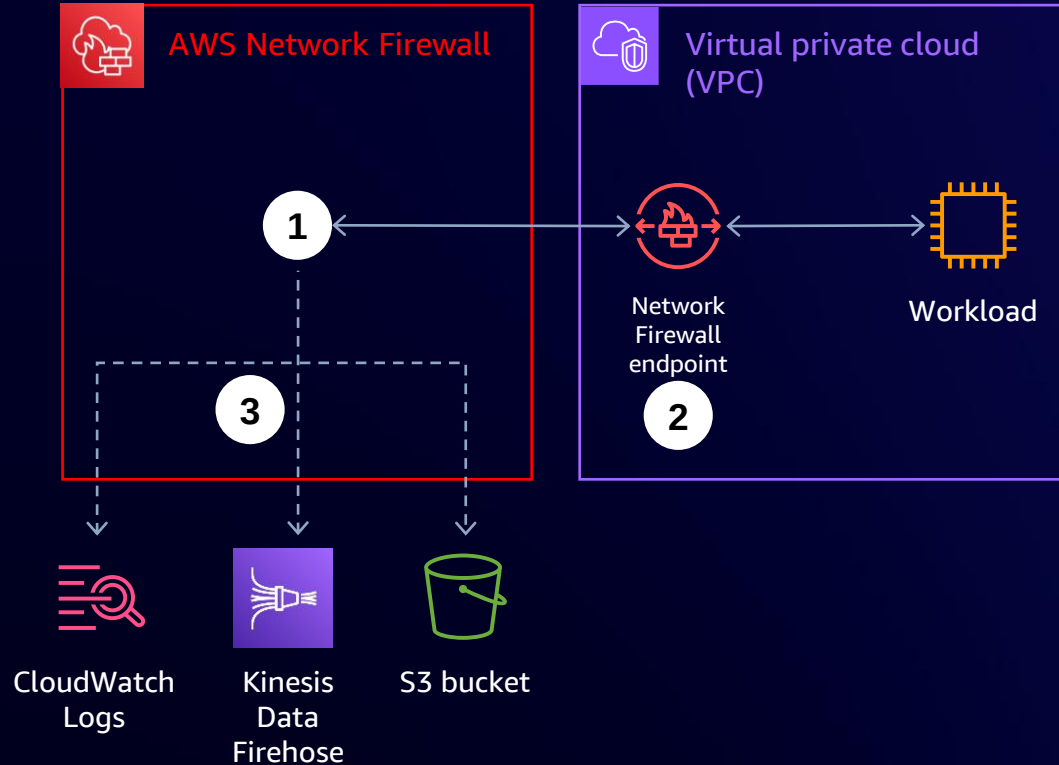
Bits per second: Varies by instance type – see <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-instance-network-bandwidth.html>

Packets per second: Varies by instance type. Provided under specific NDA, and is always given as a “up to” number – different traffic flows that are more expensive can reduce this capacity substantially

Packets per second from an instance to link-local/VPC+2, per interface: 1,024 pps ^(h)

¹ `linklocal_allowance_exceeded` is usually triggered by more than 1,024 PPS of traffic to the Route 53 DNS Resolver. This allowance is per network interface.

Monitoring AWS Network Firewall



Metrics for AWS Network Firewall are in `AWS/NetworkFirewall` namespace

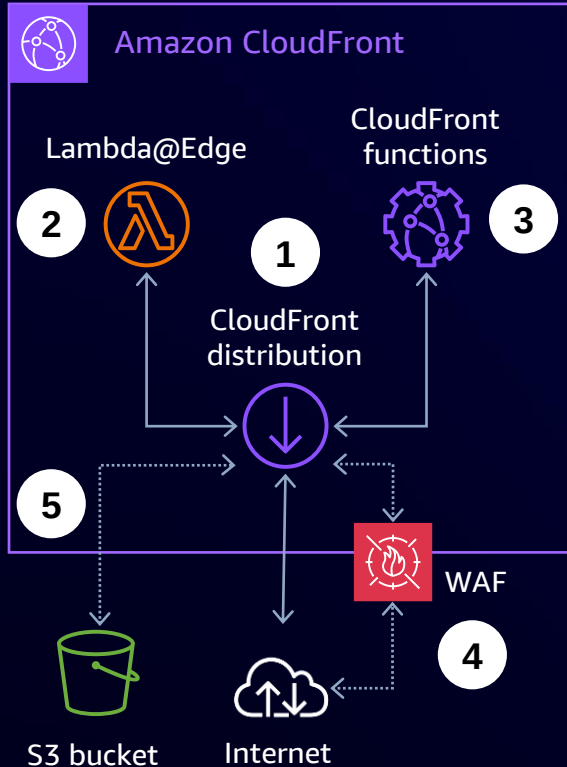
1. Alarm on `InvalidDroppedPackets` or `OtherDroppedPackets` increasing significantly; monitor `DroppedPackets`, `RejectedPackets`, and `ReceivedPackets`
2. A Firewall Endpoint is a PrivateLink service – see the PrivateLink page for monitoring details
3. Consider exporting logs from AWS Network Firewall's stateful engine.

Documentation: <https://docs.aws.amazon.com/network-firewall/latest/developerguide/monitoring-cloudwatch.html>

AWS Network Firewall key quotas:

Bits per second, per endpoint: 5 Gbps cold, able to scale up to 100 Gbps per AZ ^(h)

Monitoring Amazon CloudFront



Note that CloudWatch metrics for CloudFront (namespace `AWS/CloudFront`) are in the us-east-1 (N. Virginia) region

1. For distributions, **monitor/Alarm** `Requests` for sudden, expected changes. **Monitor/Alarm** on distributions' `TotalErrorRate`; **monitor/Alarm** on `Requests`, `BytesDownloaded`, and `BytesUploaded` **approaching** quota
2. **Monitor/Alarm** on `LambdaExecutionError`, `LambdaValidationError`, and `LambdaLimitExceededError`
3. **Monitor/Alarm** on `FunctionValidationErrors`, `FunctionExecutionErrors`, and `FunctionThrottles`; **monitor/Alarm** `FunctionComputeUtilization` **approaching** 100
4. If AWS Web Application Firewall (WAF) is in use, see the WAF slide
5. Consider enabling access logs

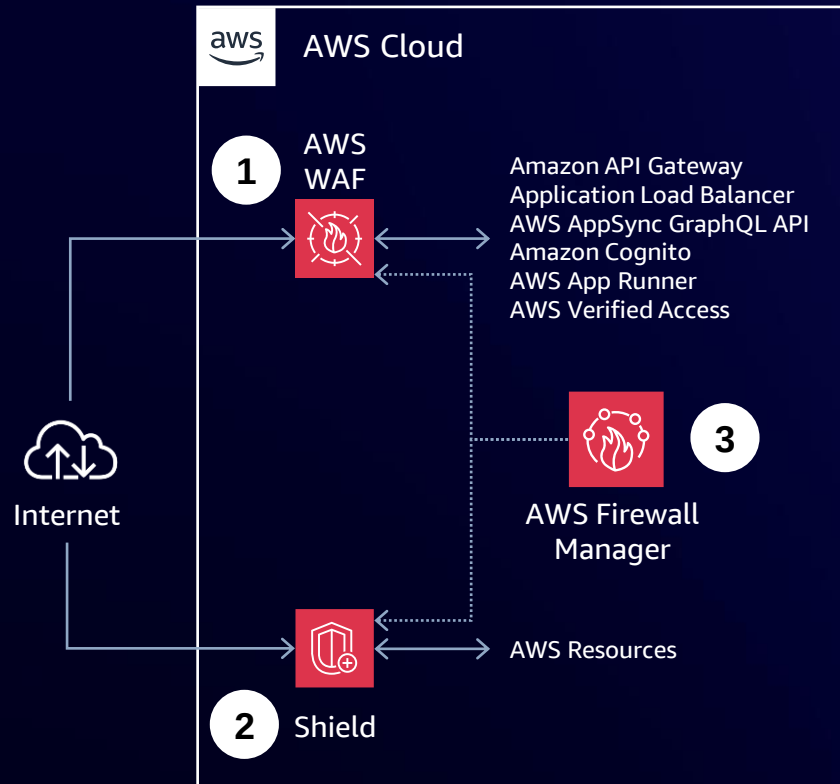
Documentation: <https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/monitoring-using-cloudwatch.html>

CloudFront key quotas:

Data transfer rate per distribution: 150 Gbps ^(s)

Requests per second, per distribution: 250,000 ^(s)

Monitoring AWS WAF and AWS Shield



1. For WAF (namespace `AWS/WAFV2`), **monitor/alarm** on `BlockedRequests` growing quickly; **monitor/alarm** on `AllowedRequests` + `BlockedRequests` + `PassedRequests` against the RPS quota
2. AWS Shield Standard is enabled by default, and transparent; consider enabling AWS Shield Advanced for high visibility websites or other resources prone to frequent DDoS attacks; if you have Advanced (namespace `AWS/DDoSProtection`), **monitor/alarm** `DDoSDetected = 1` (event is underway), and **monitor** `DDoSAttackBitsPerSecond`, `DDoSAttackPacketsPerSecond`, and `DDoSAttackRequestsPerSecond`
3. Consider enabling AWS Firewall Manager to manage AWS WAF, Shield, VPC security groups, AWS Network Firewall, and Amazon Route 53 Resolver DNS Firewall in one place

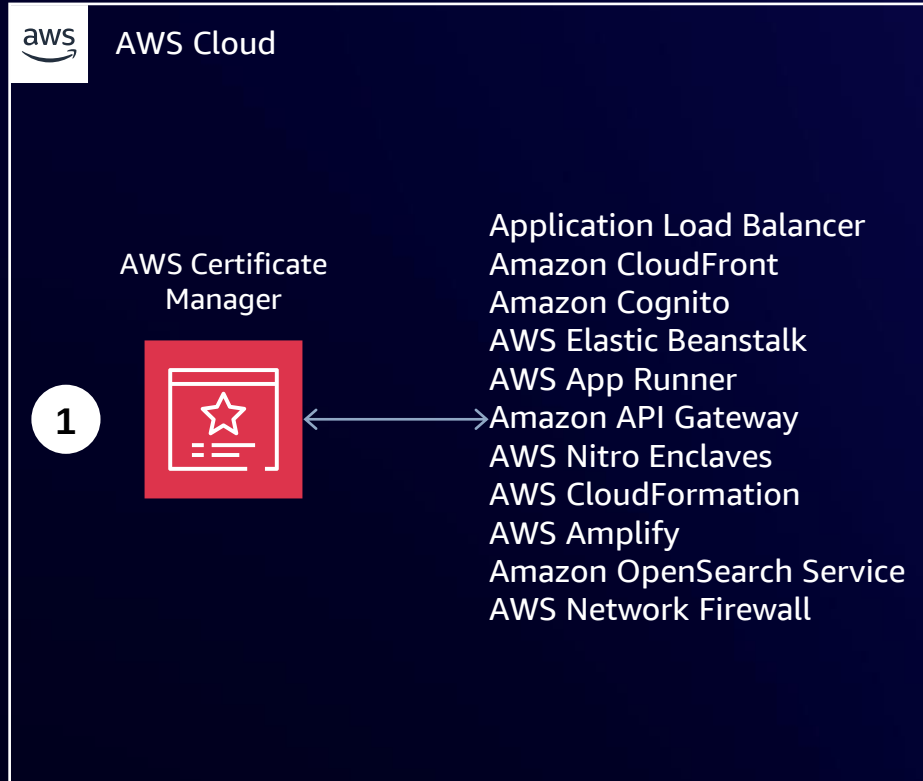
Documentation: <https://docs.aws.amazon.com/waf/latest/developerguide/monitoring-cloudwatch.html>
<https://docs.aws.amazon.com/waf/latest/developerguide/fms-chapter.html>

AWS WAF key quotas:

Requests per second, per web ACL: 25,000 ^(s)

There are no key quotas for AWS Shield Advanced

Monitoring AWS Certificate Manager



Metrics for AWS Certificate Manager are in `AWS/CertificateManager` namespace

1. Alarm on `DaysToExpiry` for each certificate, based on business rules for when renewal is required

Documentation: <https://docs.aws.amazon.com/acm/latest/userguide/cloudwatch-metrics.html>

AWS Certificate Manager:

No key quotas for AWS Certificate Manager

How traffic flows from a Linux application to the wire

Application



How traffic flows from a Linux application to the wire

Application



Nitro



How traffic flows from a Linux application to the wire

Application



Nitro

Physical network



How traffic flows from a Linux application to the wire

Application

Linux queueing discipline (qdisc) multiqueue (mq)



Nitro

Physical network



How traffic flows from a Linux application to the wire

Application

Linux queueing discipline (qdisc) multiqueue (mq)

Linux queueing discipline (qdisc) default per-queue FIFO (pfifo_fast)



Nitro

Physical network

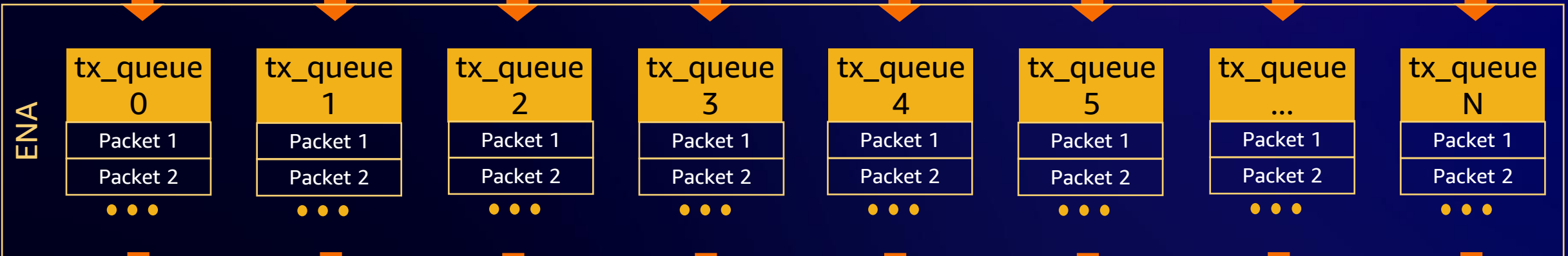


How traffic flows from a Linux application to the wire

Application

Linux queueing discipline (qdisc) multiqueue (mq)

Linux queueing discipline (qdisc) default per-queue FIFO (pfifo_fast)



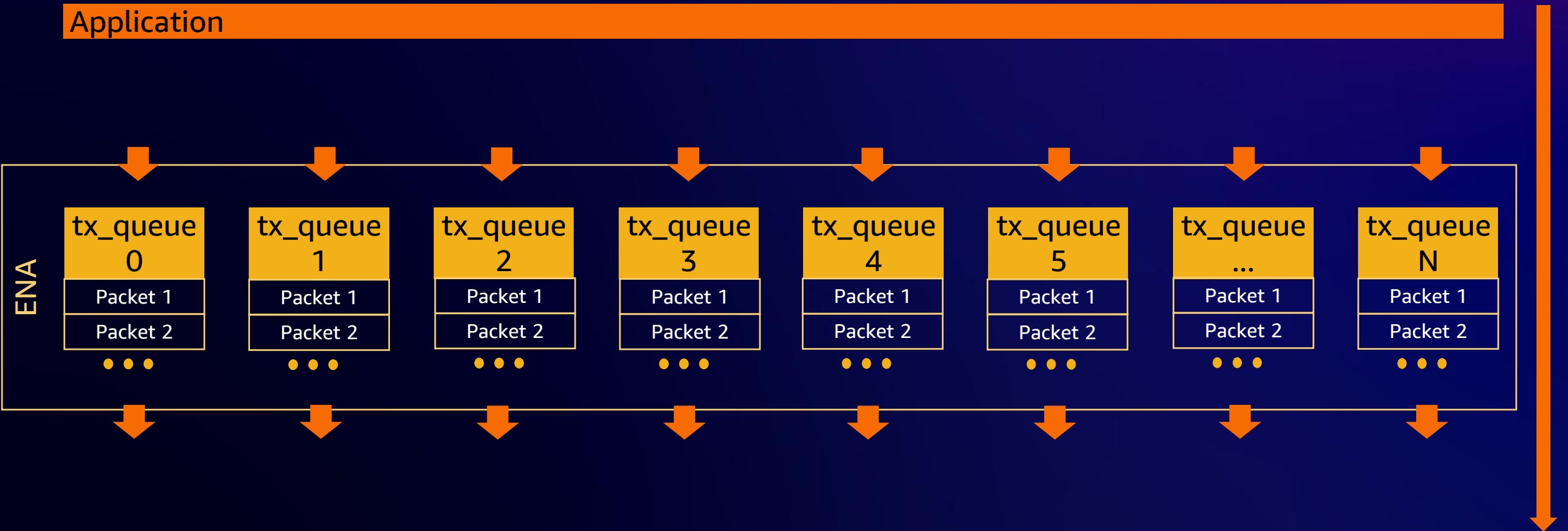
Nitro

Physical network

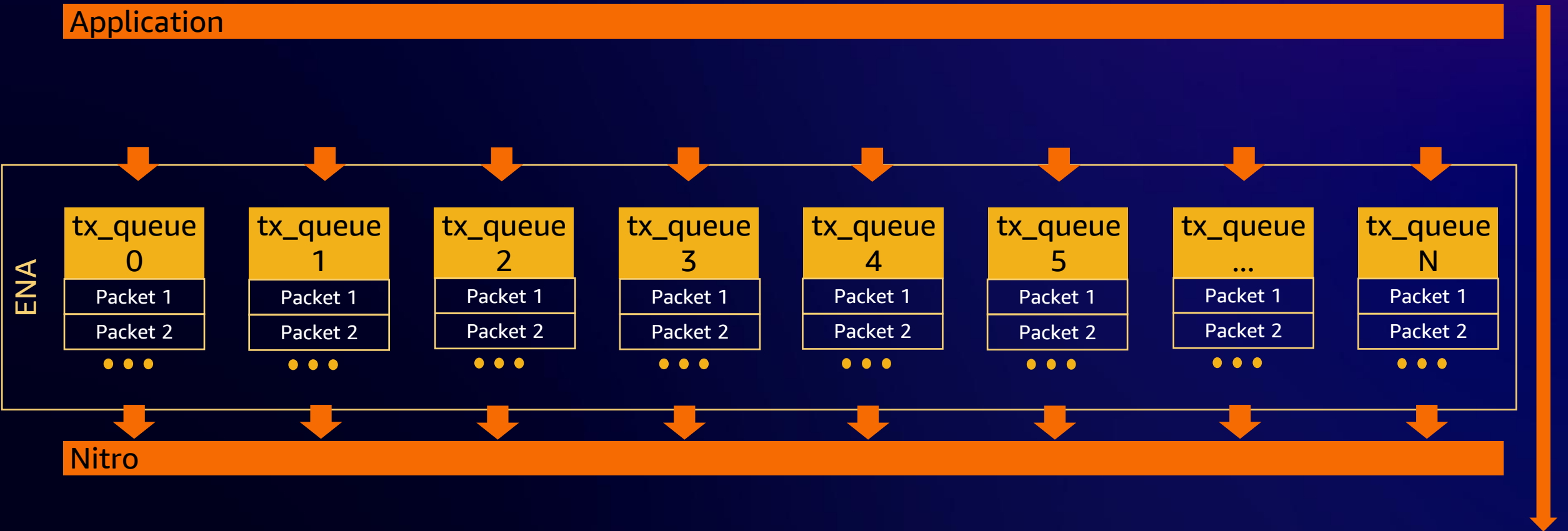
How to monitor the different layers

Application

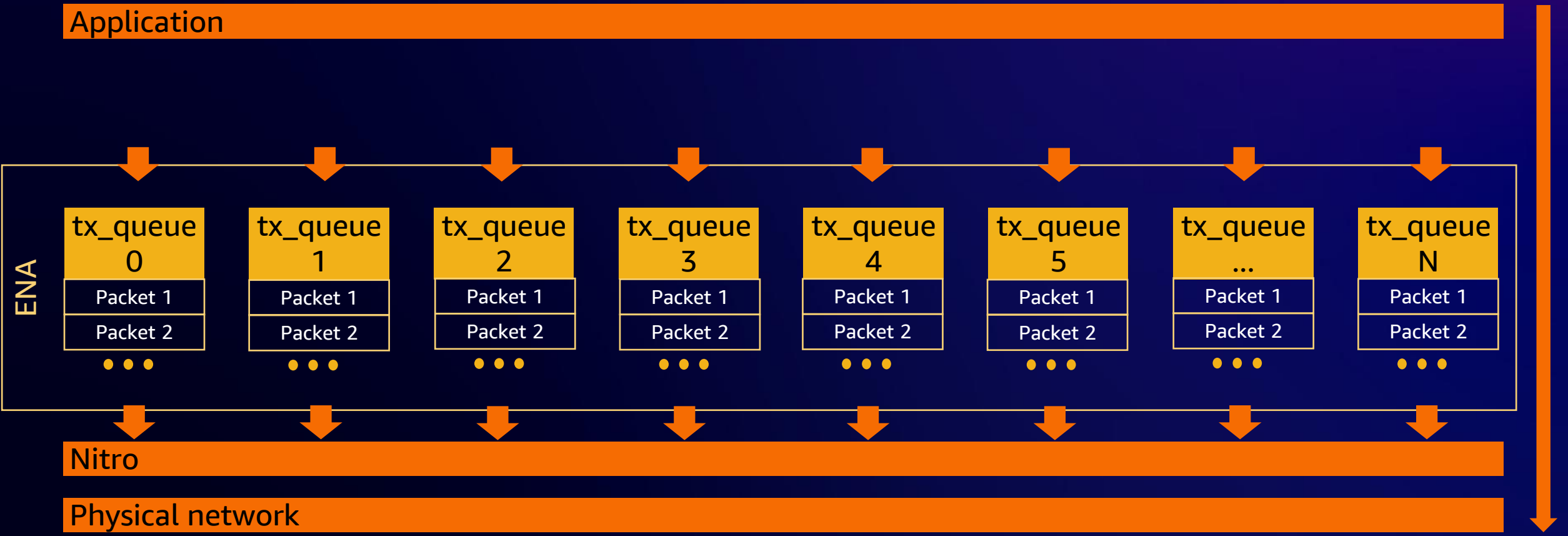
How to monitor the different layers



How to monitor the different layers



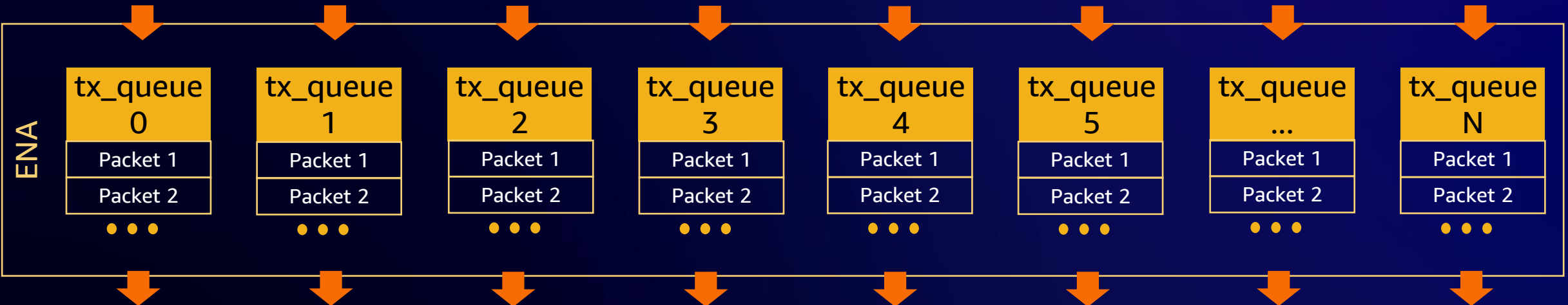
How to monitor the different layers



How to monitor the different layers

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Linux queueing discipline (qdisc) multiqueue (mq)



Nitro

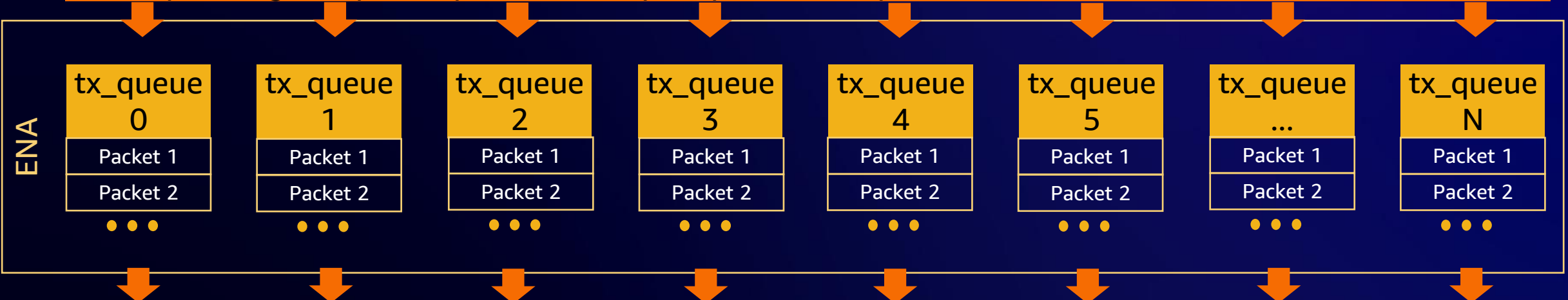
Physical network

How to monitor the different layers

Application

Linux queueing discipline (qdisc) multiqueue (mq)

Linux queueing discipline (qdisc) default per-queue FIFO (pfifo_fast)



Nitro

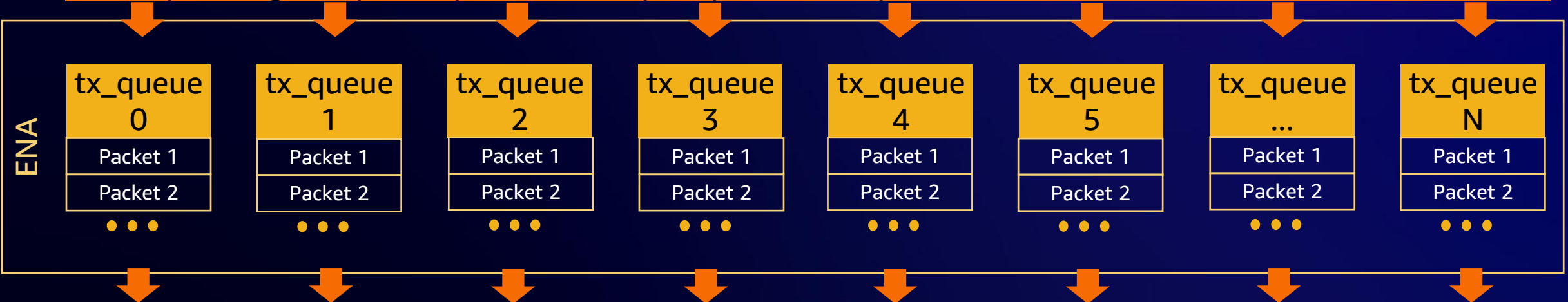
Physical network

How to monitor the different layers

Application Application-specific metrics

Linux queueing discipline (qdisc) multiqueue (mq)

Linux queueing discipline (qdisc) default per-queue FIFO (pfifo_fast)



Nitro

Physical network

How to monitor the different layers

Application

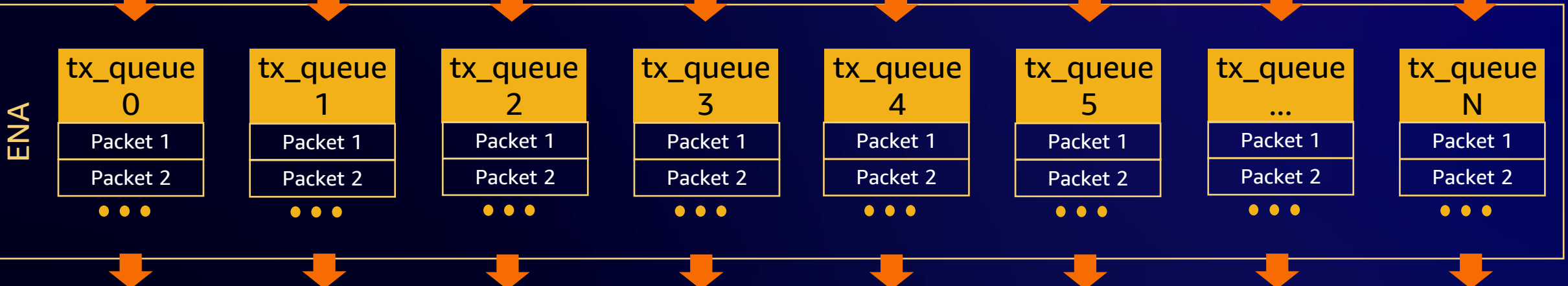
Application-specific metrics

Linux queueing discipline (qdisc) multiqueue (mq)

```
tc -s qdisc show dev eth0
```

(requires tc package to be installed)

Linux queueing discipline (qdisc) default per-queue FIFO (pfifo_fast)



Nitro

Physical network

How to monitor the different layers

Application

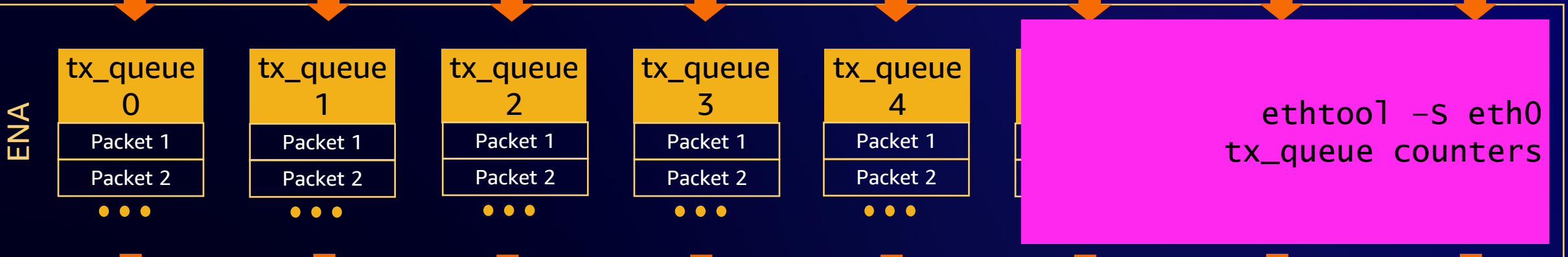
Application-specific metrics

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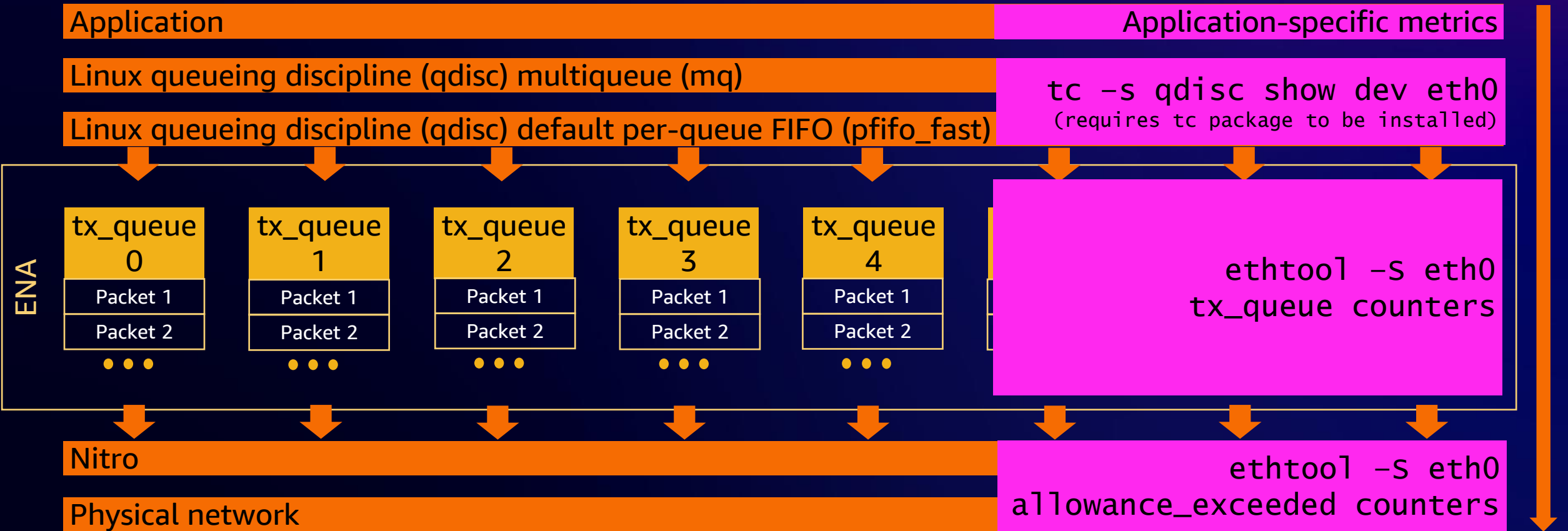
Linux queueing discipline (qdisc) default per-queue FIFO (pfifo_fast)



Nitro

Physical network

How to monitor the different layers



What do customers need to monitor for RX?

```
# ethtool -S eth0 | grep -E "exceed|contrack"
```

```
bw_in_allowance_exceeded: 0  
bw_out_allowance_exceeded: 0  
pps_allowance_exceeded: 0  
contrack_allowance_exceeded: 0  
linklocal_allowance_exceeded: 0  
contrack_allowance_available: 4925267
```

contrack_allowance_available should be monitored – if this number is approaching 0, remediation should be performed

*_allowance_exceeded can be queues or drops¹ – small increments are OK, but large/constant increments are signs of problems

```
[root@ip-172-31-61-17 ec2-user]# ifconfig eth0
```

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9001  
inet 10.10.0.25 netmask 255.255.255.0 broadcast 10.10.0.255  
inet6 fe80::4da:2fff:fe92:3e4b prefixlen 64 scopeid 0x20<link>  
ether 06:da:2f:92:3e:4b txqueuelen 1000 (Ethernet)  
RX packets 42390523889 bytes 5426227831002 (4.9 TiB)  
RX errors 0 dropped 8277 overruns 0 frame 0  
TX packets 297837941 bytes 42149472220 (39.2 GiB)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

RX "dropped" indicates packets that were received, but no RX buffer was available to store the packet

What do customers need to monitor for TX?

```
# ethtool -S eth0 | grep -E "exceed|stop|contrack"
```

```
bw_in_allowance_exceeded: 0
bw_out_allowance_exceeded: 0
pps_allowance_exceeded: 0
contrack_allowance_exceeded: 0
linklocal_allowance_exceeded: 0
contrack_allowance_available: 4925267
queue_0_tx_queue_stop: 10984
queue_1_tx_queue_stop: 11055
queue_2_tx_queue_stop: 4750
queue_3_tx_queue_stop: 12998
queue_4_tx_queue_stop: 10418
queue_5_tx_queue_stop: 12938
queue_6_tx_queue_stop: 10720
queue_7_tx_queue_stop: 12168
```

*_tx_queue_stops are a backpressure mechanism – by themselves, they don't indicate drops, but increments here indicate getting very close to drops occurring

contrack_allowance_available should be monitored – if this number is approaching 0, remediation should be performed

*_allowance_exceeded can be queues or drops¹ – small increments are OK, but large/constant increments are signs of problems

```
[root@ip-172-31-61-17 ec2-user]# tc -s qdisc show dev eth0
```

```
qdisc mq 0: root
  Sent 847325121 bytes 10699805 pkt (dropped 47916229, overlimits 0 requeues 7)
  backlog 297560b 6924p requeues 7
qdisc pfifo_fast 0: parent :8 bands 3 priomap 1 2 2 2 1 2 0 0 1 1 1 1 1 1 1 1
  Sent 117164784 bytes 2724576 pkt (dropped 9456602, overlimits 0 requeues 1)
  backlog 42312b 986p requeues 1
qdisc pfifo_fast 0: parent :7 bands 3 priomap 1 2 2 2 1 2 0 0 1 1 1 1 1 1 1 1
  Sent 112173371 bytes 2607411 pkt (dropped 10037854, overlimits 0 requeues 1)
  backlog 42742b 996p requeues 1
qdisc pfifo_fast 0: parent :6 bands 3 priomap 1 2 2 2 1 2 0 0 1 1 1 1 1 1 1 1
  Sent 108425318 bytes 2521351 pkt (dropped 4012900, overlimits 0 requeues 0)
  backlog 42527b 989p requeues 0
qdisc pfifo_fast 0: parent :5 bands 3 priomap 1 2 2 2 1 2 0 0 1 1 1 1 1 1 1 1
  Sent 84451300 bytes 1963732 pkt (dropped 8111352, overlimits 0 requeues 1)
  backlog 42398b 986p requeues 1
qdisc pfifo_fast 0: parent :4 bands 3 priomap 1 2 2 2 1 2 0 0 1 1 1 1 1 1 1 1
  Sent 126486519 bytes 2941431 pkt (dropped 4034796, overlimits 0 requeues 1)
  backlog 42226b 981p requeues 1
qdisc pfifo_fast 0: parent :3 bands 3 priomap 1 2 2 2 1 2 0 0 1 1 1 1 1 1 1 1
  Sent 119635095 bytes 2782059 pkt (dropped 182659, overlimits 0 requeues 2)
  backlog 43b 2p requeues 2
qdisc pfifo_fast 0: parent :2 bands 3 priomap 1 2 2 2 1 2 0 0 1 1 1 1 1 1 1 1
  Sent 90320743 bytes 2100421 pkt (dropped 5297845, overlimits 0 requeues 0)
  backlog 42312b 982p requeues 0
qdisc pfifo_fast 0: parent :1 bands 3 priomap 1 2 2 2 1 2 0 0 1 1 1 1 1 1 1 1
  Sent 88728879 bytes 2060240 pkt (dropped 6784713, overlimits 0 requeues 1)
  backlog 42355b 985p requeues 1
```

qdisc "dropped" are packet drops.